

18.06 Linear Algebra - Lectures 28, 29, 30

Textbook Notes - Sections 6.2, 7.1, 7.2, 8.1

• Section 6.2 Similar Matrices

Two matrices that describe the same linear map in different bases are similar.

Similarity partitions all matrices into families.

Each family has a single simplest representation

↳ diagonal if possible

↳ Jordan form otherwise

• Definition Similar Matrices

A and C are similar $\Leftrightarrow C = B^{-1}AB$ for some invertible B

• Similarity preserves the following:

↳ eigenvalues (with multiplicity)

↳ trace

↳ rank

characteristic polynomial
determinant

of independ. eigenvectors

↳ Not preserved:

↳ Eigenvectors, if $Ax = \lambda x$ then $C(B^{-1}x) = \lambda(B^{-1}x)$ which has the same eigenvalue ~~by~~ but the eigenvector has been transformed by B^{-1} .

• Proof Eigenvalues Match:

$$\begin{aligned}\det(C - \lambda I) &= \det(B^{-1}AB - \lambda B^{-1}B) = \det(B^{-1}) \det(A - \lambda I) \det(B) \\ &= \det(A - \lambda I)\end{aligned}$$

• Diagonalizable case

When A has n independent eigenvectors, choose $B = X$ (eigenvector matrix):

$$X^{-1} A X = \Lambda$$

↳ Λ is the simplest member of the family

↳ Every matrix similar to Λ is diagonalizable with those eigenvalues.

• Jordan Form (general case):

When eigenvectors are missing, the family contains no diagonal matrix.

↳ Best available representation $\rightarrow$ Jordan Form.

$$B^{-1} A B = J = \text{diag}(J_1, J_2, \dots, J_s)$$

↳ With one "block" per independent eigenvalue vector.

↳ Each Block has a single eigenvalue on the diagonal and 1's directly above:

$$J_i = \begin{bmatrix} \lambda_i & 1 & 0 \\ 0 & \lambda_i & 1 \\ 0 & 0 & \lambda_i \end{bmatrix}$$

• Key Facts:

- ↳ s blocks $\Leftrightarrow s$ independent eigenvectors
- ↳ one off-diagonal 1 for each missing eigenvector.
- ↳ all blocks $1 \times 1 \Leftrightarrow J = \Lambda \Leftrightarrow$ diagonalizable
- ↳ Two matrices are similar if and only if they share the same Jordan Form

Generalized Eigenvectors:

↳ The columns of B beyond the true eigenvectors satisfy
 $(A - \lambda I)b_{k+1} = b_k$ (each chain to eigenvector not zero)

Use:

↳ $\frac{du}{dt} = Au$ becomes $u(t) = Be^{Jt}B^{-1}u(0)$

↳ Since J is triangular, e^{Jt} produces $e^{\lambda t}$ times powers $1, t, \dots, t^{s-1}$ (where $te^{\lambda t}$ originates)

Practical Caveat:

↳ Jordan form numerically unstable

↳ Any perturbation separates repeated eigenvalues & removes the off-diagonal 1 's

↳ Theoretical not computational tool.

◦ Section 7.1 Image Processing by LA

An image is a matrix of greyscale values.

Nearby pixels are correlated, so matrix is close to low rank.

Low Rank matrices compress enormously.

• The Compression Argument

An $m \times n$ matrix has mn entries. A single rank-one piece UV^T needs only $m+n$ numbers.

Keeping the first K rank-one pieces costs $K(m+n+1)$ instead of mn .

→ Rank 1 Example:

a 6×6 matrix of constant $g \rightarrow g \cdot (\text{col of 1s}) \cdot (\text{row of 1s})$
36 numbers $\rightarrow$ 12 numbers (or 1, with ones vector predefined).

• SVD's Contribution:

$$A = \sigma_1 U_1 V_1^T + \sigma_2 U_2 V_2^T + \dots + \sigma_r U_r V_r^T$$

With $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, the pieces arrive in order of importance. Truncating after K terms gives the best, possible rank- K approximation to A .

• Edges (sudden greyscale jumps) raise ^{of} ~~rank~~ effective rank.
Cartoons compress worse than photographs. →

• Adaptive vs Preselected Bases

↳ The Fourier basis is fixed in advance, so the FFT is fast.

↳ The SVD builds its basis from the image itself → better compression; higher cost.

Section 7.2 Bases & Matrices in SVD :

Diagonalization $A = X\Lambda X^{-1}$ fails in three ways

1. eigenvectors are usually not orthogonal
2. there may not be enough eigenvectors
3. A must be square.

The SVD fixes all three by using two orthonormal bases instead of one.

SVD Factorization

$$A = U\Sigma V^T$$

- $\rightarrow V$ is $n \times n$ orthogonal (right singular vectors)
- $\rightarrow U$ is $m \times m$ orthogonal (left singular vectors)
- $\rightarrow \Sigma$ is $m \times n$ diagonal with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$

Defining Relation :

$$Av_i = \sigma_i \cdot u_i \quad \text{for } i = 1, \dots, r$$

A maps each $\underline{v}$ in the row space to a multiple of the corresponding $\underline{u}$ in the column space.

$\hookrightarrow$ "Diagonalization with two bases"

• Bases for all four subspace

$U_1, \dots, U_r$	orthonormal basis for $C(A)$	column space
$U_{r+1}, \dots, U_m$	orthonormal basis for $N(A^T)$	left nullspace
$V_1, \dots, V_r$	orthonormal basis for $C(A^T)$	row space
$V_{r+1}, \dots, V_n$	orthonormal basis for $N(A)$	nullspace

→ the Fundamental Theorem of Linear Algebra with orthonormal bases throughout.

• How to compute it:

1. From $A^T A$ (symmetric positive semidefinite)

↳ Its orthonormal eigenvectors are the V 's

2. Its eigenvalues are σ_i^2

↳ Take positive square roots for the σ 's, ordered descending.

3. Get each U from $U_i = \frac{A V_i}{\sigma_i}$, $i \leq r$

4. Extend with nullspace bases to make U and V square

• Why it works?

Substituting $A = U \Sigma V^T$

$$\rightarrow A^T A = V \Sigma^T \Sigma V^T$$

→ V 's are eigenvectors of $A^T A$

→ eigenvalues σ_i^2

$$\rightarrow A A^T = U \Sigma \Sigma^T U^T$$

→ U 's are eigenvectors of $A A^T$

→ same eigenvalues, non-zero

• Rank Argument + Tips

$$\text{trace}(A^T A) = \text{trace}(A A^T) = \sum \sigma_i^2$$

If $A^T A$ is 3×3 with A ~~2×3~~ 2×3

↳ then $r \leq \min(2, 3) = 2$

↳ so $A^T A$ being 3×3 means $3 - 2 = 1$ zero eigenvalue

Key Step

The U 's come out orthogonal automatically

$$\begin{aligned} U_i^T U_j &= (A v_i)^T (A v_j) / (\sigma_i \sigma_j) = v_i^T (A^T A) v_j / (\sigma_i \sigma_j) \\ &= \sigma_j^2 (v_i^T v_j) / (\sigma_i \sigma_j) = \underline{0} \quad \text{for } i \neq j \end{aligned}$$

Orthogonality of the $\underline{V}$'s (guaranteed, since $A^T A$ symmetric) forces orthogonality of the $\underline{U}$'s.

Rank-one Expansion

$$A = \sigma_1 u_1 v_1^T + \dots + \sigma_r u_r v_r^T$$

Truncating after k terms gives the optimal rank- k approximation (Eckart-Young) $\rightarrow$ the compression result from 7.1.

When SVD equals diagonalization

$A = U \Sigma V^T$ coincides with $A = X \Lambda X^{-1}$ exactly when A is symmetric positive semidefinite.

Then $U = V = Q$ and $\Sigma = \Lambda$

For symmetric A with negative eigenvalues, $\sigma_i = |\lambda_i|$ and the sign is absorbed into $\underline{U}$.

Singular Values vs Eigenvalues

$$\sigma_i \geq |\lambda|_{\max} \quad \text{and} \quad \sigma_n \leq |\lambda|_{\min} \quad (\text{for square } A)$$

$$\sigma_1 \sigma_2 \dots \sigma_n = |\det A|$$

Singular values are always real and nonnegative, even when eigenvalues are complex.

◦ Section 8.1 Linear Transformations:

A matrix is one representation of something more general:
a map T that preserves linear combinations.

Stating the idea without coordinates clarifies what matrices do.
It extends the whole theory to derivatives, integrals, and function spaces.

• Linear Transform Definition

$$T(cv + dw) = cT(v) + dT(w) \quad \text{for all } v, w \text{ \& all scalars } c, d$$

↳ Equivalently: $T(v + w) = T(v) + T(w)$
 $T(cv) = c \cdot T(v)$

↳ Consequence: $T(0) = 0$

• Recognizing Linearity

Linear

Not Linear

- $T(v) = Av$
- dot product $T(v) = a \cdot v$
- rotation, projection, reflection
- derivative $T(f) = \frac{df}{dx}$
- integral $T(f) = \int f$

- $T(v) = \|v\|$ (triangle inequality, fails for $c < 0$)
- $T(v) = v + u_0$ (shift, fails $T(0) = 0$)
- anything with v_i^2 , $v_1 v_2$, or lengths

* $T(v) = Av + u_0$ is called Affine $\rightarrow$ straight lines stay straight, but it is not linear.

The Basis determines everything

If $T(v_1), \dots, T(v_n)$ are known for a basis, then $T(u)$ is known for every u .

Since $u = c_1 v_1 + \dots + c_n v_n$, linearity gives

$$T(u) = c_1 T(v_1) + \dots + c_n T(v_n)$$

↳ Why matrices work: the columns of A are exactly T applied to the standard basis vectors.

Geometric Effect

- Linear transformations send lines to lines and preserve equal spacing.

- The midpoint $\frac{1}{2}(v+w)$ maps to $\frac{1}{2}(T(v) + T(w))$.

- Triangles map to triangles with proportions intact, and the origin stays fixed.

Example: the Derivative as a Matrix

Input space = quadratics $\{1, x, x^2\}$ (dimension 3)

Output space = linear functions $\{1, x\}$ (dimension 2).

$$T(1) = 0 \quad T(x) = 1 \quad T(x^2) = 2x$$

Writing outputs in the output basis gives the columns:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

The nullspace of T is the constants (dim 1). The $C(A)$ is all of $\{1, x\}$ (dim 2). The Counting Theorem still holds:

$$\hookrightarrow \dim(C(A)) + \dim(N(A)) = 2 + 1 = 3 = \dim(\text{input space})$$

This is why the pseudoinverse appears: the integral $T'(f) = \int_0^x f(t) dt$ satisfies $TT' = I$ but not $T'T = I$ (integrating after differentiating loses constants).

- Composition

$$(ST)(v) = S(T(v))$$

The composition of linear transformations is linear, and it corresponds to matrix multiplication.

↳ this is the conceptual reason matrix multiplication is defined the way it is.

- Subspace Equivalent

Range → The set of all possible outputs.
Feed in every vector and collect what comes out.

Kernel → The set of inputs that map to zero.